

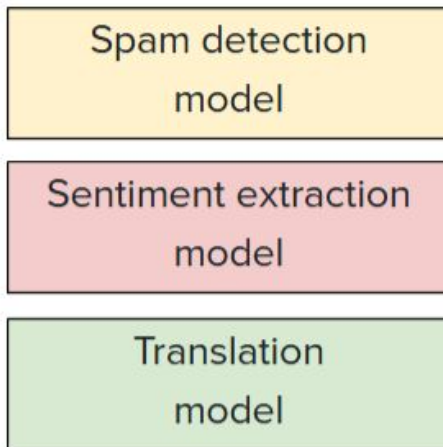
LLMs Training and Tuning

Deep Learning

Aziz Temirkhanov
Lambda, HSE

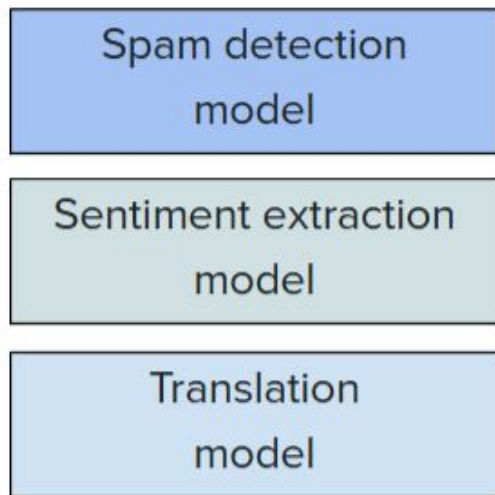
Model Training

Traditional ML. Given task, train model from scratch



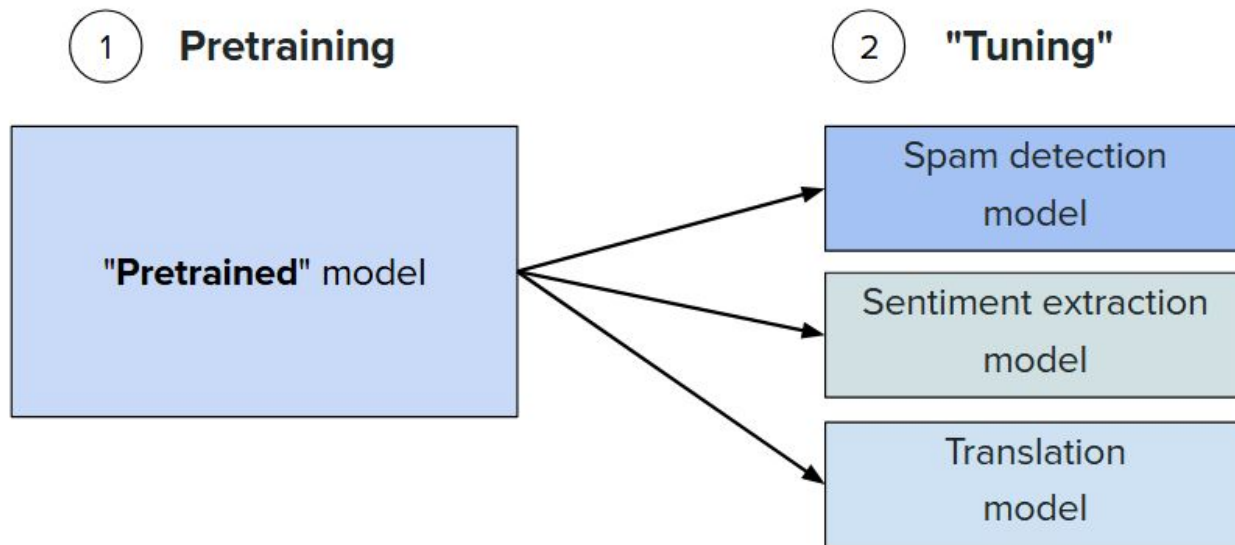
Model Training

Transfer learning. Reuse trained model to some capacity



LLM Training

LLM training. Train LLM to understand language, then tune it for end task



LLM Training

Goal. Learn pattern of language and code

English

My cute teddy bear is reading by the window, his tiny paws holding a book almost bigger than him. The afternoon light dances across his soft fur as he turns each page with great care...

Multi-lingual

Mon ours en peluche est mon plus fidèle compagnon. Toujours assis sur mon lit, il m'attend patiemment, prêt à écouter mes secrets...

خرس تدي من صميمي ترين دوست من هست. همیشه رو تخت من نشسته و حاضره به قصه های من گوش بده...

...

teddy_bear.py

```
class TeddyBear:
    def __init__(self, name):
        self.name = name

    def hug(self):
        print(f"{self.name} gives a warm hug!")

TeddyBear("Fluffy").hug()...
```

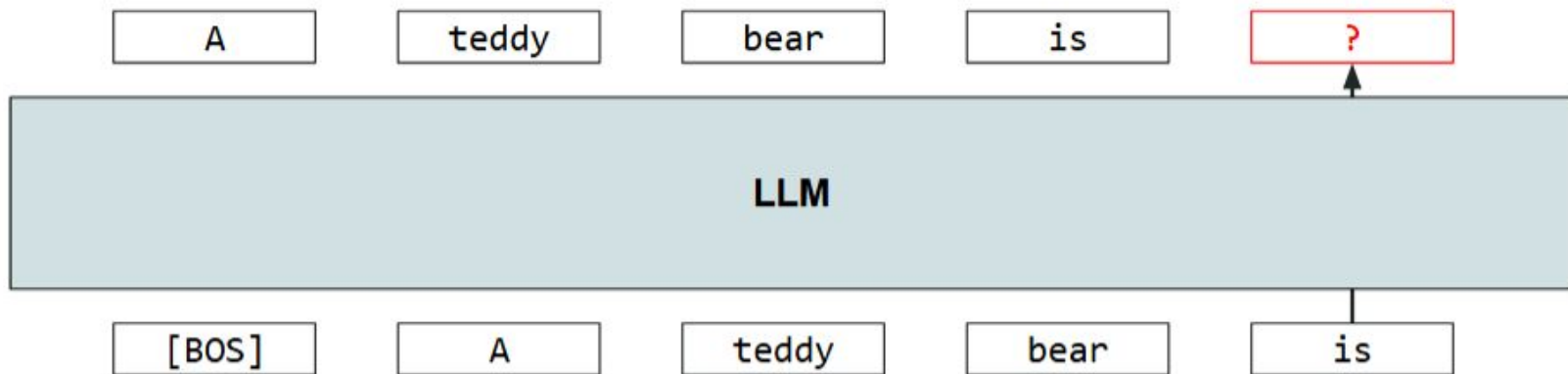
teddy_bear.go

```
package main

func main(){
    println("Teddy bear hug!")
}
```

LLM Training

Objective function. Predict next token



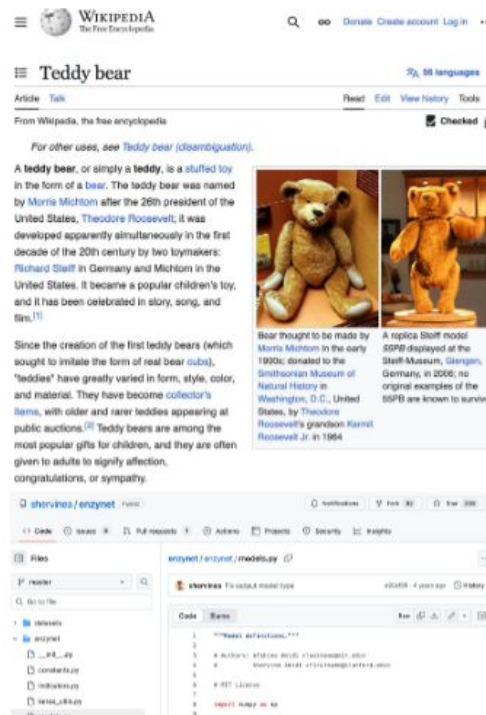
LLM Training

Goal. Learn pattern of language and code

Objective function. Predict next token

Data mixtures.

- Web-scraped (e.g. Common Crawl, Wikipedia)
- Code (GitHub, StackOverflow)



FLOPs and FLOPS

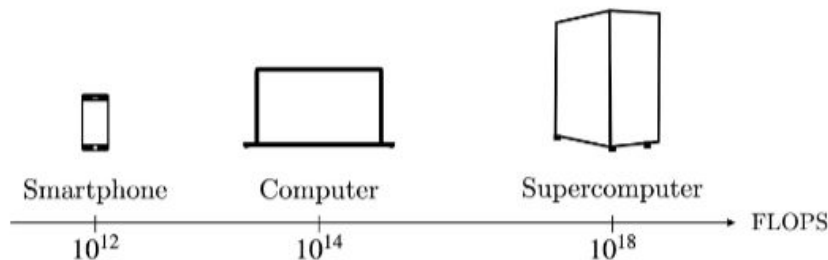
 **FLOPs**
Floating-point **O**perations

 **FLOPS** or **FLOP/s**
Floating-point **O**perations
per **S**econd

$f(\text{\#tokens}, \text{\#parameters})$

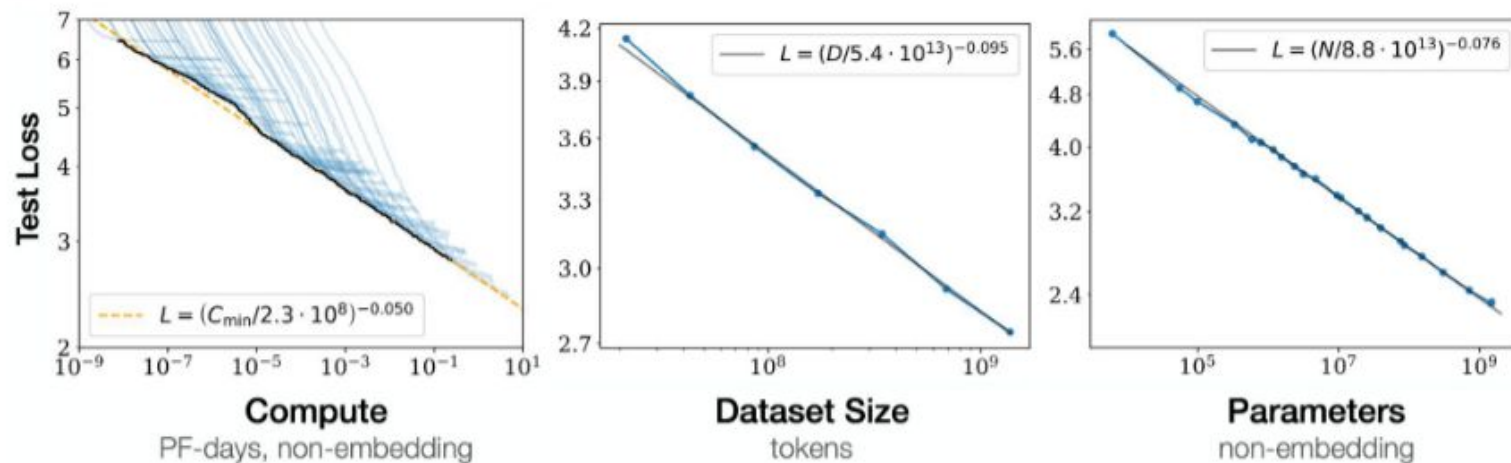


Computational performance



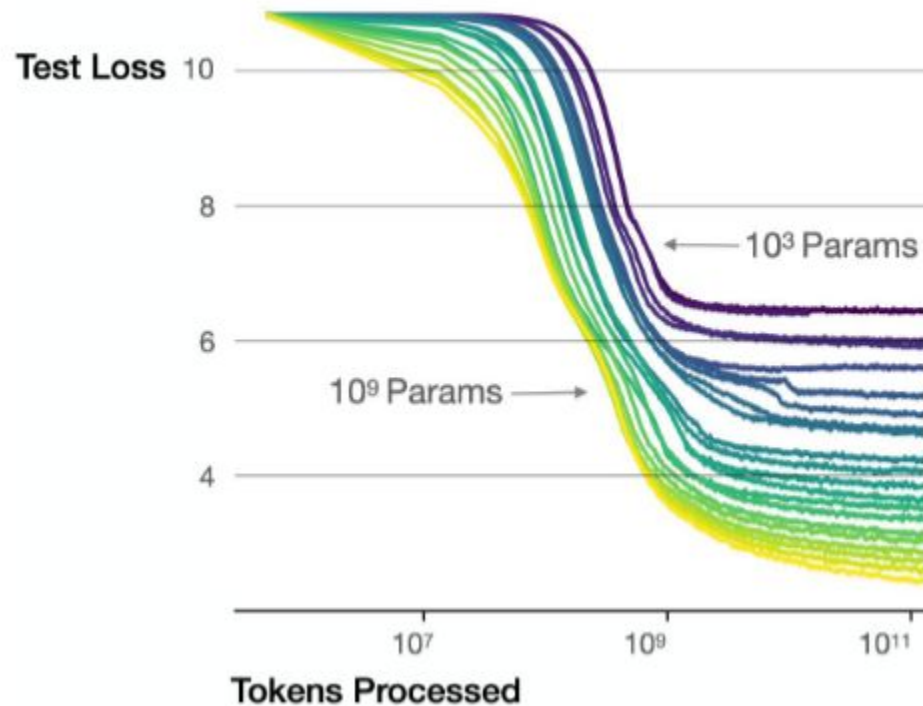
Scaling Laws

Scaling.



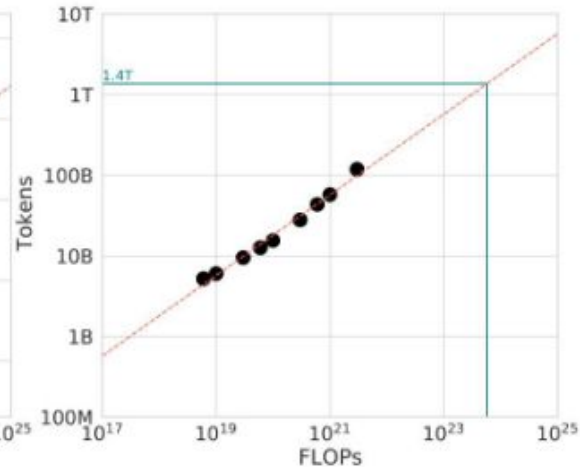
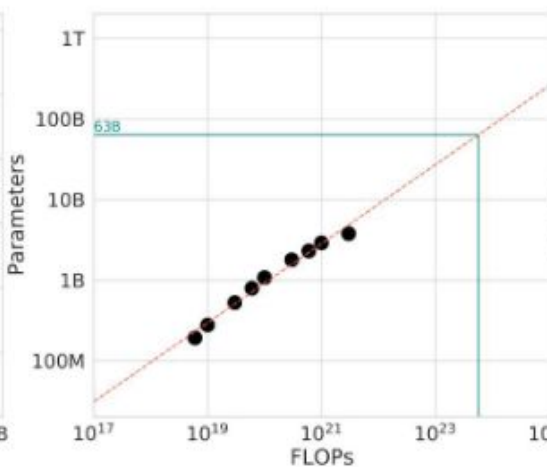
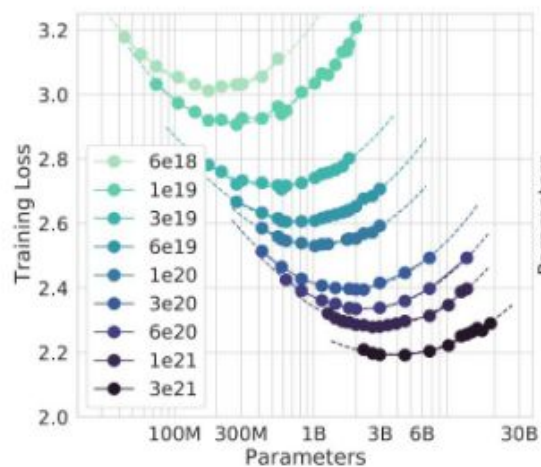
Scaling Laws

Sample efficiency.



Scaling Laws

Chinchilla law.



Scaling Laws

Chinchilla law.

Parameters	FLOPs	FLOPs (in <i>Gopher</i> unit)	Tokens
400 Million	1.92e+19	1/29,968	8.0 Billion
1 Billion	1.21e+20	1/4,761	20.2 Billion
10 Billion	1.23e+22	1/46	205.1 Billion
67 Billion	5.76e+23	1	1.5 Trillion
175 Billion	3.85e+24	6.7	3.7 Trillion
280 Billion	9.90e+24	17.2	5.9 Trillion
520 Billion	3.43e+25	59.5	11.0 Trillion
1 Trillion	1.27e+26	221.3	21.2 Trillion
10 Trillion	1.30e+28	22515.9	216.2 Trillion

Problems

Challenges

Cost.

- (At least) millions of dollars
- Takes time
- Environment / electricity

Learned knowledge.

- "Knowledge cutoff"
- Hard to edit knowledge
- "Plagiarism"

The screenshot displays the OpenAI GPT-5 interface. At the top, the 'GPT-5' model is selected with a 'Default' dropdown and a share icon. Below this, a row of metrics is shown: REASONING (4 dots, 'Higher'), SPEED (3 lightning bolts, 'Medium'), PRICE (\$1.25 · \$10, 'Input · Output'), INPUT (Text, image), and OUTPUT (Text). A descriptive paragraph states: 'GPT-5 is our flagship model for coding, reasoning, and agentic tasks across domains. Learn more in our [GPT-5 usage guide](#).' To the right, a list of features is provided: '400,000 context window', '128,000 max output tokens', 'Sep 30, 2024 knowledge cutoff' (highlighted with a red box), and 'Reasoning token support'.

GPT-5 Default

The best model for coding and agentic tasks across domains

Compare Try in Playground

REASONING	SPEED	PRICE	INPUT	OUTPUT
Higher	Medium	\$1.25 · \$10 Input · Output	Text, image	Text

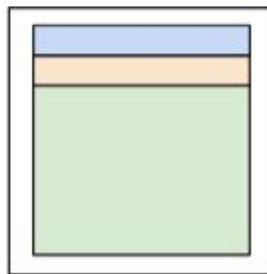
GPT-5 is our flagship model for coding, reasoning, and agentic tasks across domains. Learn more in our [GPT-5 usage guide](#).

- ✦ 400,000 context window
- ↔ 128,000 max output tokens
- Sep 30, 2024 knowledge cutoff
- 💡 Reasoning token support

Optimization. Data Parallelism

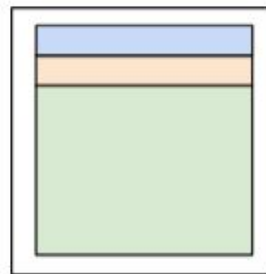
Idea.

- Divide batch of data across devices
- Model replicated on each device



GPU 1

...



GPU n

parameters

gradients

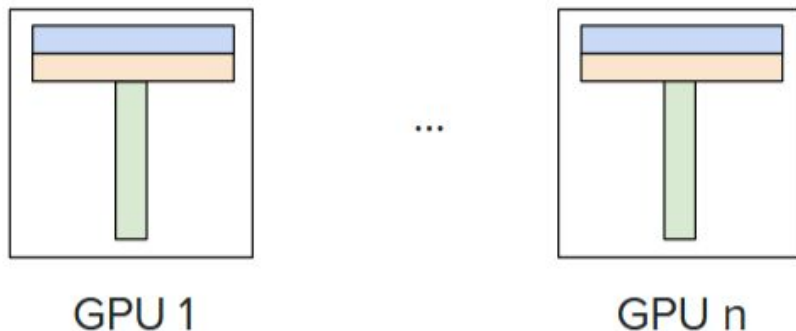
optimizer state

Data Parallelism

Idea. Redundant information across devices

ZeRO = **Z**ero **R**edundancy **O**ptimization

Variations. ZeRO-1: Share **optimizer state**

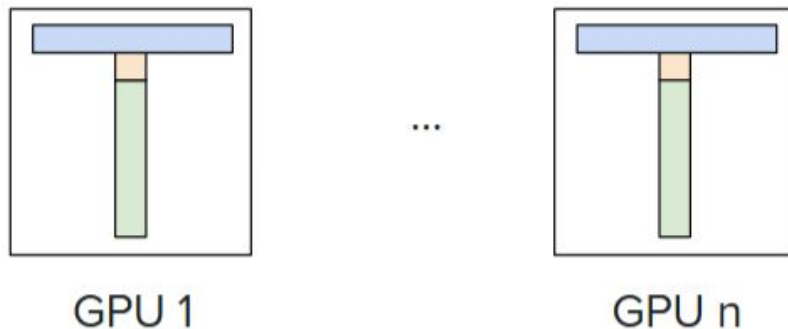


Data Parallelism

Idea. Redundant information across devices

ZeRO = **Z**ero **R**edundancy **O**ptimization

Variations. ZeRO-2: Share **optimizer state** + **gradients**



Data Parallelism

Idea. Redundant information across devices

ZeRO = **Z**ero **R**edundancy **O**ptimization

Variations. ZeRO-3: Share **optimizer state** + **gradients** + **parameters**



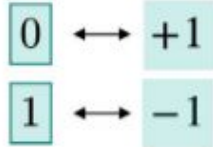
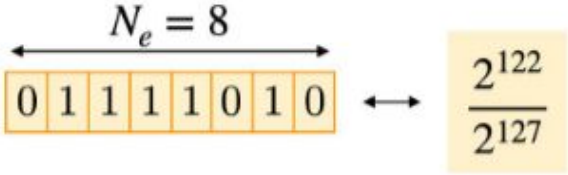
Model Parallelism

Idea. Split the model computations across several devices

Variations.

- **Tensor Parallelism (TP).**
- **Pipeline Parallelism (PP).**
- **Sequence Parallelism (SP).**
- **Context Parallelism (CP).**
- **Expert Parallelism (EP).**

Floating Point

Name	Description	Illustration
Sign	Controls whether the number is positive or negative. Typically takes up to 1 bit.	 0 ↔ +1 1 ↔ -1
Exponent	Controls the magnitude of the number. Also called <i>range</i> .	 $N_e = 8$ 0 1 1 1 1 0 1 0 ↔ $\frac{2^{122}}{2^{127}}$
Mantissa	Controls the granularity of the number, i.e. what is after the decimal point. Also called <i>significand</i> or <i>fraction</i> .	 1 1 0 0 0 ... ↔ 1.75

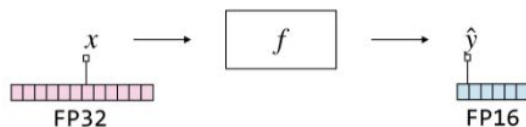
Floating Point

	Sign	Exponent	Mantissa
FP16 (Floating-Point 16)	1	5	10
FP32 (Floating-Point 32)	1	8	23
FP64 (Floating-Point 64)	1	11	52
BFLOAT16 (Brain Float 16)	1	8	7

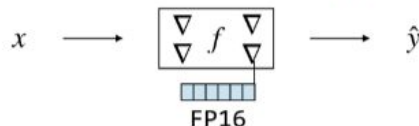
Mixed Precision

Objective. Speed up training and decrease memory requirements

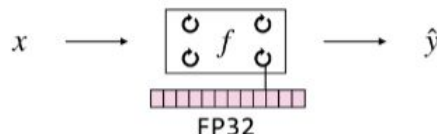
- **Forward pass.** Activations in low precision



- **Backwards pass.** Gradient updates in low precision

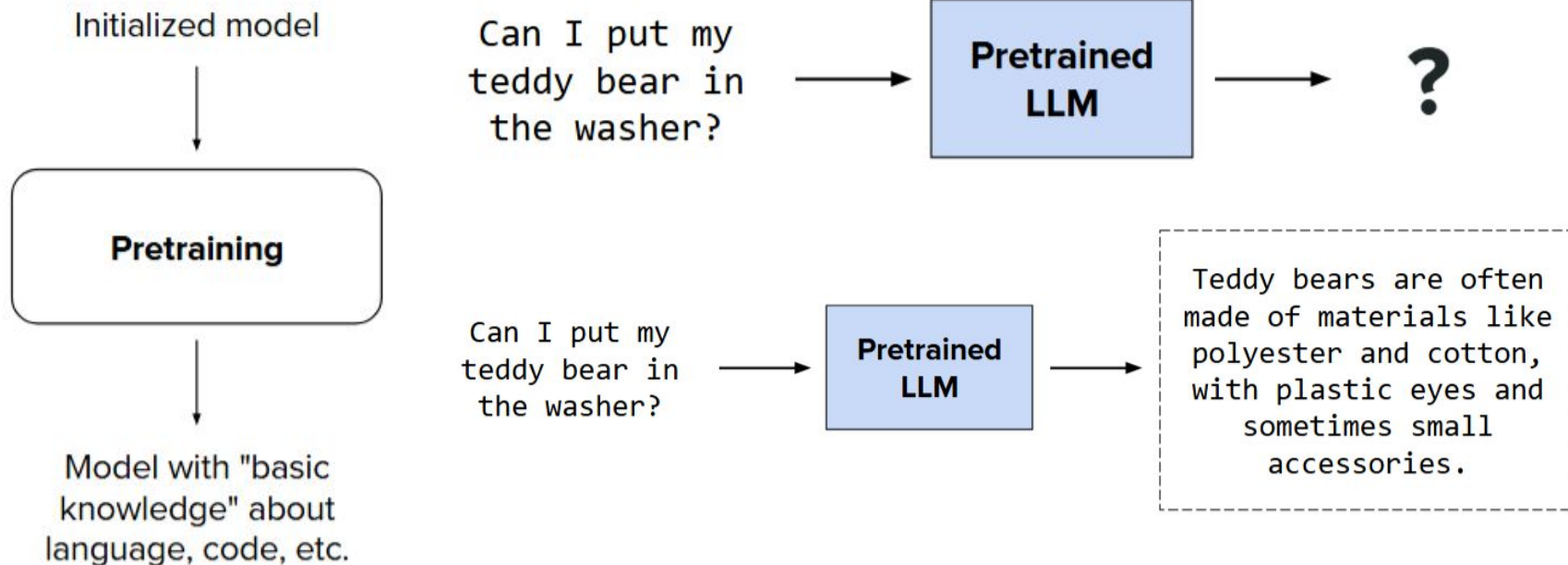


- **Weights update.** Keep weights in high precision

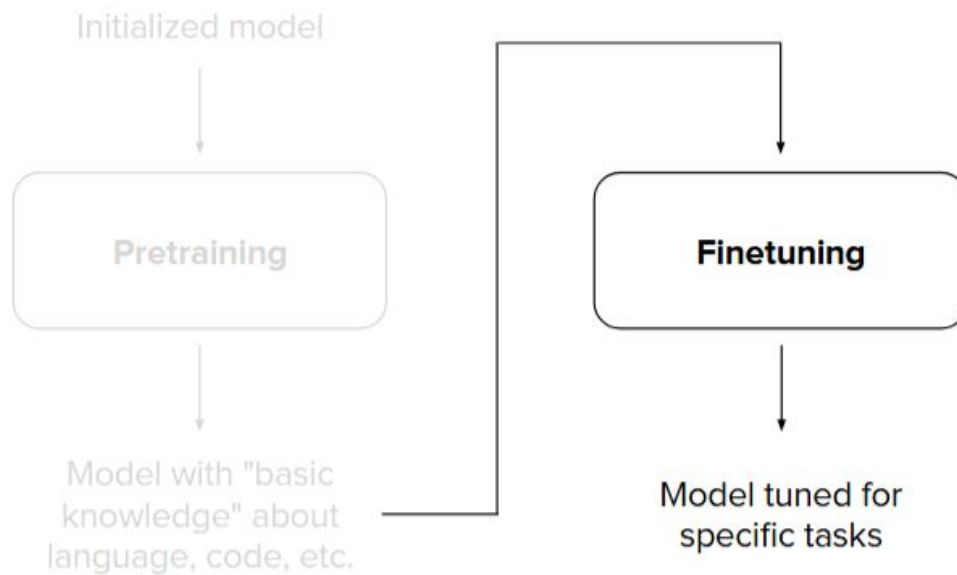


Fine Tuning

Why Fine Tune?



Solution



Supervised Finetuning

SFT = Supervised FineTuning

Idea. Change the model behavior by **tuning its weights**

Strategy.

- Collect pairs of input/outputs with desired behavior (aka SFT data)
- Train using next word prediction objective *given the input*

Special case. SFT on instruction-following data: "Instruction tuning"

Instruction Tuning

Goal. "Graduate" the model to being a helpful assistant.

Story writing

Write a short story about a teddy bear who likes to read poetry.

Once upon a time, a bear, Teddy, stumbled upon verses from Attar. The bear became so captivated with the rhythm of the poem's sentences that he got filled with happiness.

Poem creation

Create a poem about my cute teddy bear.

Soft and cuddly, full of charm,
Always keeps me safe from harm,
With button eyes and fur so neat,
My teddy bear is oh so sweet.

Lists generation

List three fun activities a teddy bear might do on a rainy day.

Sure!

1. Read poetry with friends.
2. Be cute.
3. Hug its owner tightly.

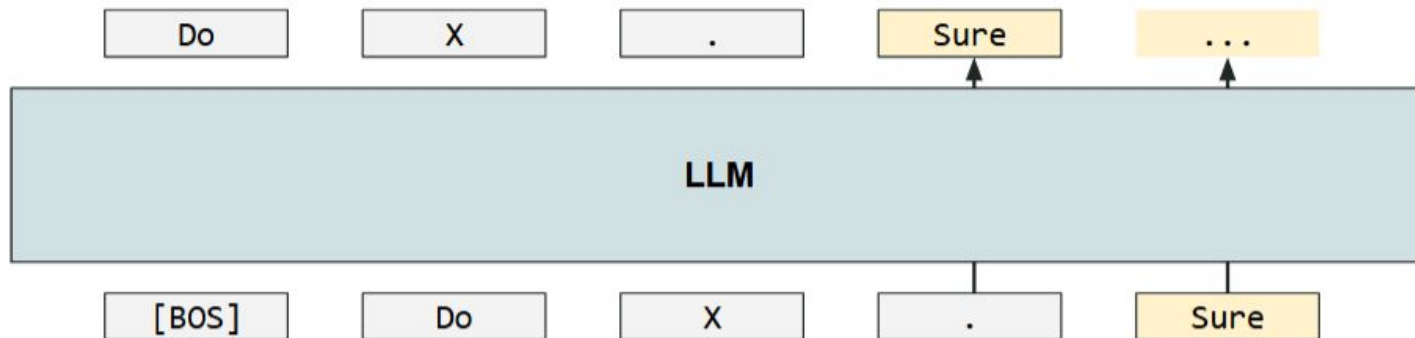
Explanation

Explain why a teddy bear is a great friend.

A teddy bear is a great friend because it provides comfort and companionship. Its soft fur and cuddly nature is perfect for hugs. Teddy bears are always there in both happy and sad moments.

Instruction Tuning

Objective function. Predict next token *given the input*.

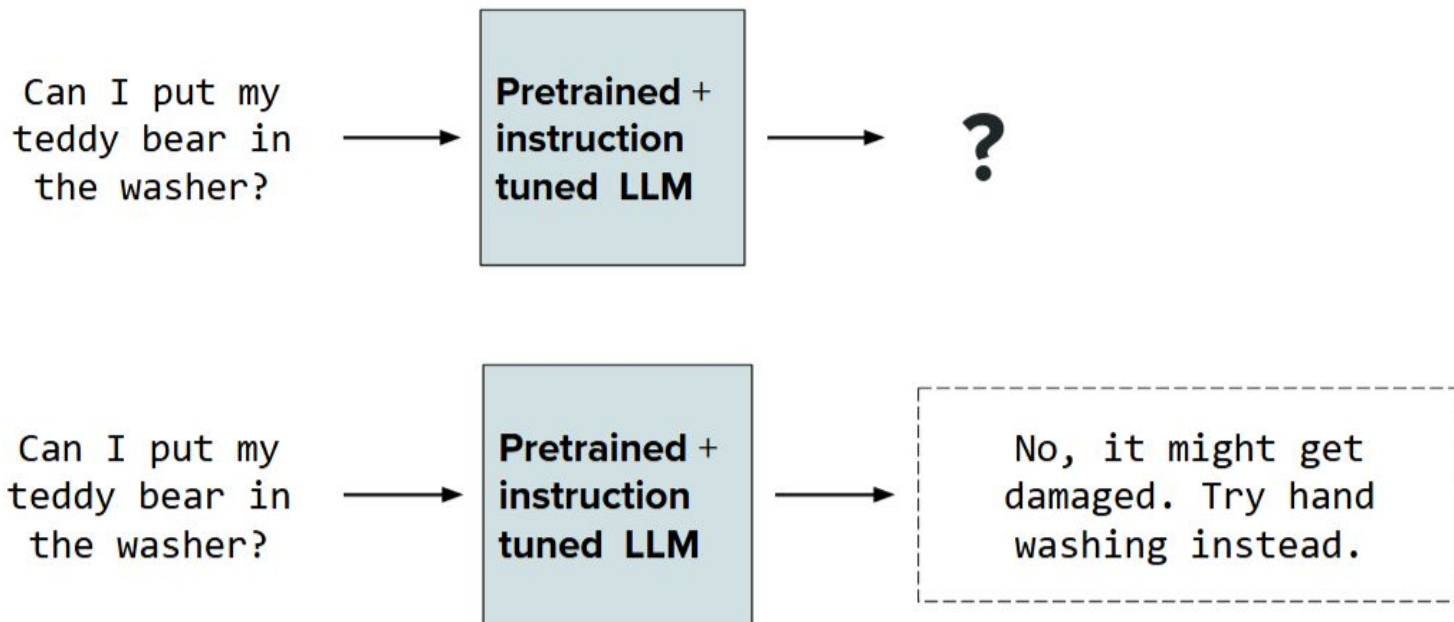


Instruction Tuning

Data mixtures. Can be both human-written and synthetic data.

- Assistant dialogs
- Synthetic instructions
- Maths, reasoning, code
- Safety alignment
- ...

Result



Challenges

- Very **high-quality** data needed
- Sensitive to **prompt distribution**
- **Generalization**
- Difficult to **evaluate**
- Computationally **expensive**

Benchmarks

Dimensions.

- General knowledge: MMLU
- Basic reasoning: ARC-Challenge
- Math reasoning: GSM8K
- Code generation: HumanEval

Validity. Recommended to train on the test *task* to compare across models.